

# Finding the global minimum

The 2-D Ackley function using a Genetic Algorithm

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SUBMITTED TO

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# Find the lowest point

## The task

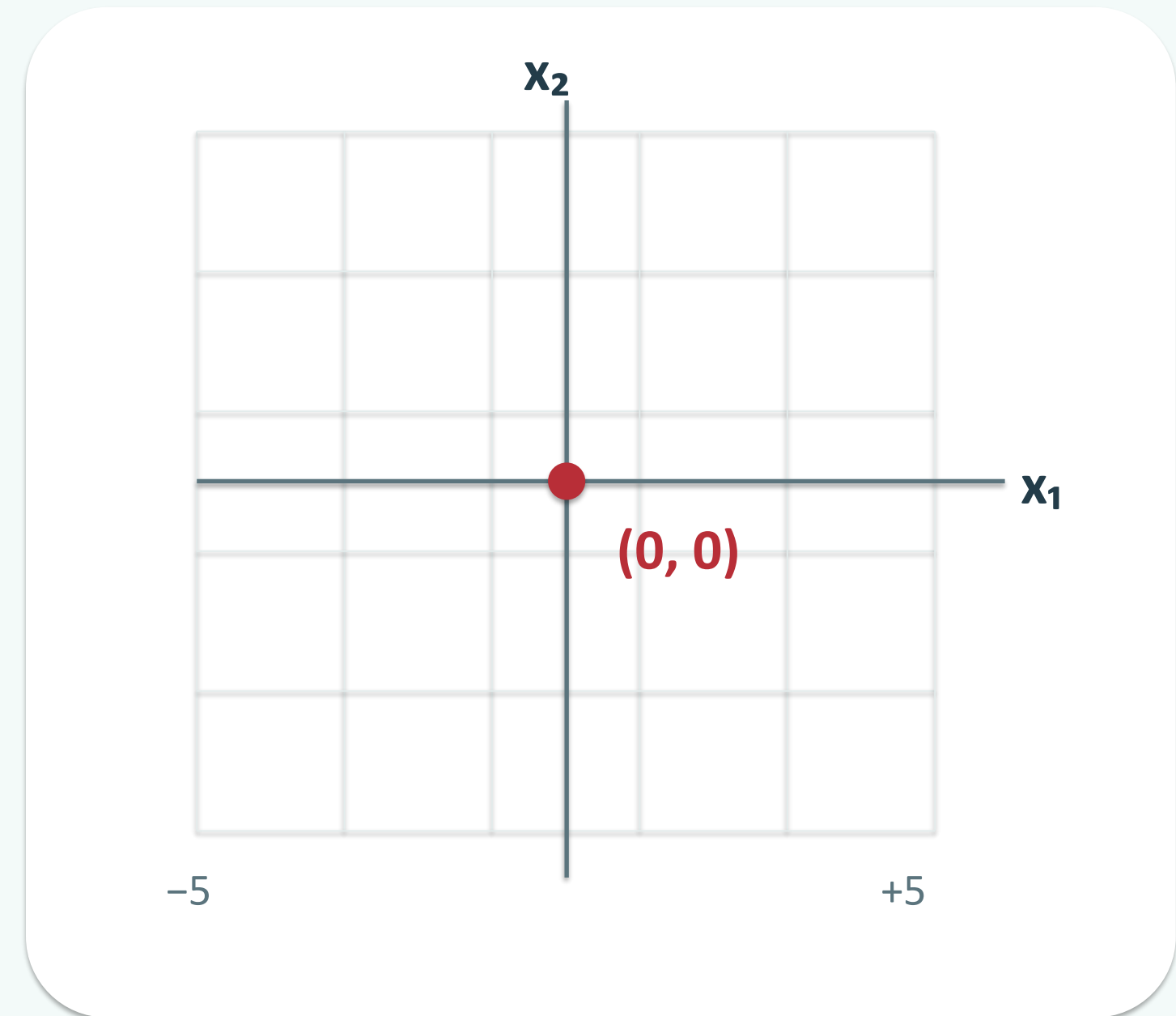
Choose  $x_1$  and  $x_2$  to make  $f$  as small as possible.

## The search range

Both coordinates stay between  $-5$  and  $+5$ .

## The known answer

$x^* = (0, 0)$  and  $f(x^*) = 0$



The search evaluates candidate points; it is not given the answer coordinates.

# The Ackley function in 2-D

START WITH  $n$  VARIABLES

$$f(\mathbf{x}) = -a \exp\left(-b \sqrt{\frac{1}{n} \sum_{i=1}^n x_i^2}\right) - \exp\left(\frac{1}{n} \sum_{i=1}^n \cos(cx_i)\right) + a + e$$

Set  $n = 2$



Expand the two sums



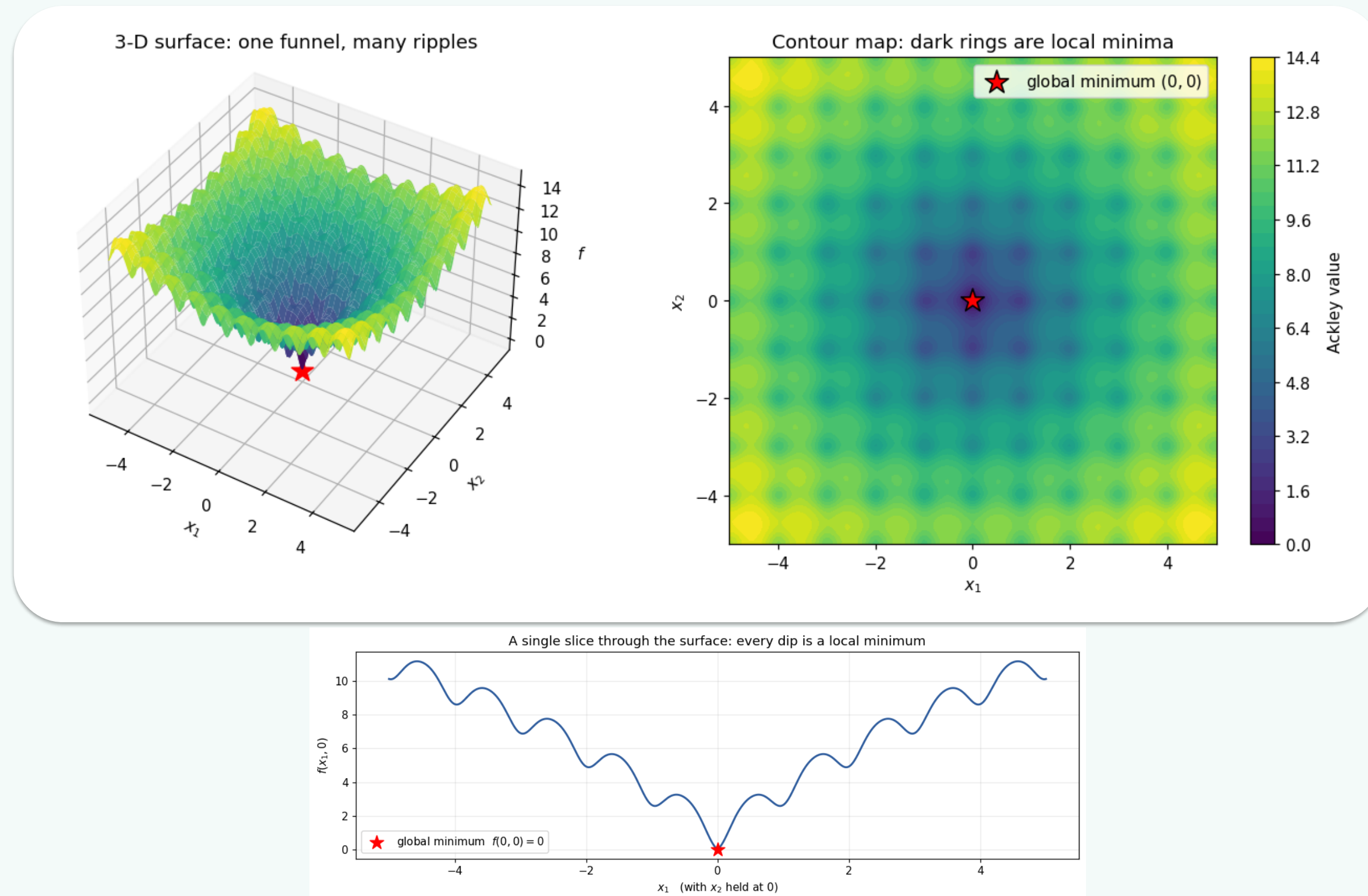
Use  $a=20$ ,  $b=0.2$ ,  $c=2\pi$

THE FUNCTION IN OUR CODE

$$f(x_1, x_2) = -20 \exp\left(-0.2 \sqrt{\frac{x_1^2 + x_2^2}{2}}\right) - \exp\left(\frac{\cos(2\pi x_1) + \cos(2\pi x_2)}{2}\right) + 20 + e$$

At the origin:  $-20 - e + 20 + e = 0$

# One deep minimum. Many small traps.



## A wide funnel

The smooth part points toward the centre.

## Many local dips

The cosine part adds small traps.

## Expensive to scan

A  $10^{-6}$  grid needs about  $10^{14}$  points.

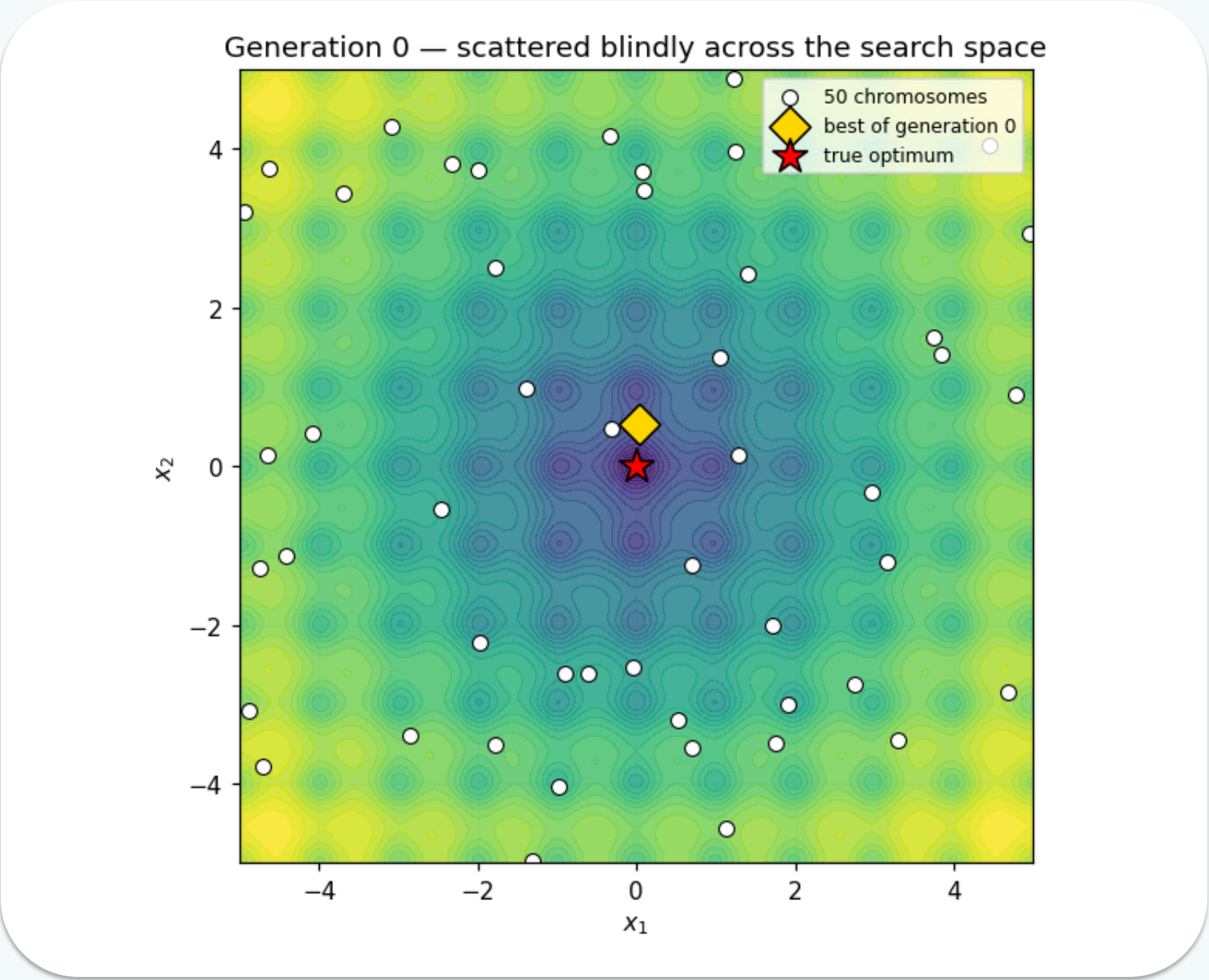
**A downhill method can stop in a local minimum.**

# Search with a population

BIOLOGY → OUR PROGRAM

| Term       | Meaning               |
|------------|-----------------------|
| Population | 50 candidate points   |
| Chromosome | One pair $[x_1, x_2]$ |
| Gene       | One coordinate        |
| Fitness    | Selection weight      |

Evaluate → select → mix  
→ mutate → repeat



A population explores many points. Better candidates get more chances to reproduce.

# One solution = two real numbers

## VALUE ENCODING

GENE 1

 $X_1$ 

GENE 2

 $X_2$ 

One chromosome

**Example A**

[-4.0, 2.5]

**Example B**

[-1.5, -1.0]

Each gene stays in [-5, +5]. Real-valued encoding still has finite computer precision.

# The settings we used

ASSIGNMENT SETTINGS

| Parameter        | Value       |
|------------------|-------------|
| Population       | 50          |
| Max. generations | 100         |
| Crossover        | 80%         |
| Mutation         | 5% per gene |
| Elitism          | 1           |

Value encoding • roulette selection  
One-point crossover

OUR IMPLEMENTATION CHOICES

| Choice               | Value                                         |
|----------------------|-----------------------------------------------|
| Mutation type        | Gaussian                                      |
| Step size $\sigma$   | $\text{clip}(f_{\text{best}}/4, 10^{-12}, 2)$ |
| Exact-copy limit     | 15 of 50                                      |
| Tolerance / patience | $10^{-10}$ / 20 rounds                        |
| Main random seed     | 7                                             |

Probability is fixed;  
step size adapts to the result.

# The search loop

## 1 Initialise

50 random points → evaluate → sort



## 2 Check stopping rules

If yes: report best and finish

NO: CONTINUE

## 4 Make 50 children

Crossover → mutate → clip → evaluate



## 3 Set $\sigma$ and probabilities

Then select 50 parents by roulette

## 5 Choose the next 50

1 elite + 49 seats, with the copy limit



## 6 Repeat

Use the survivors as the next population

↻ Return to the stopping check

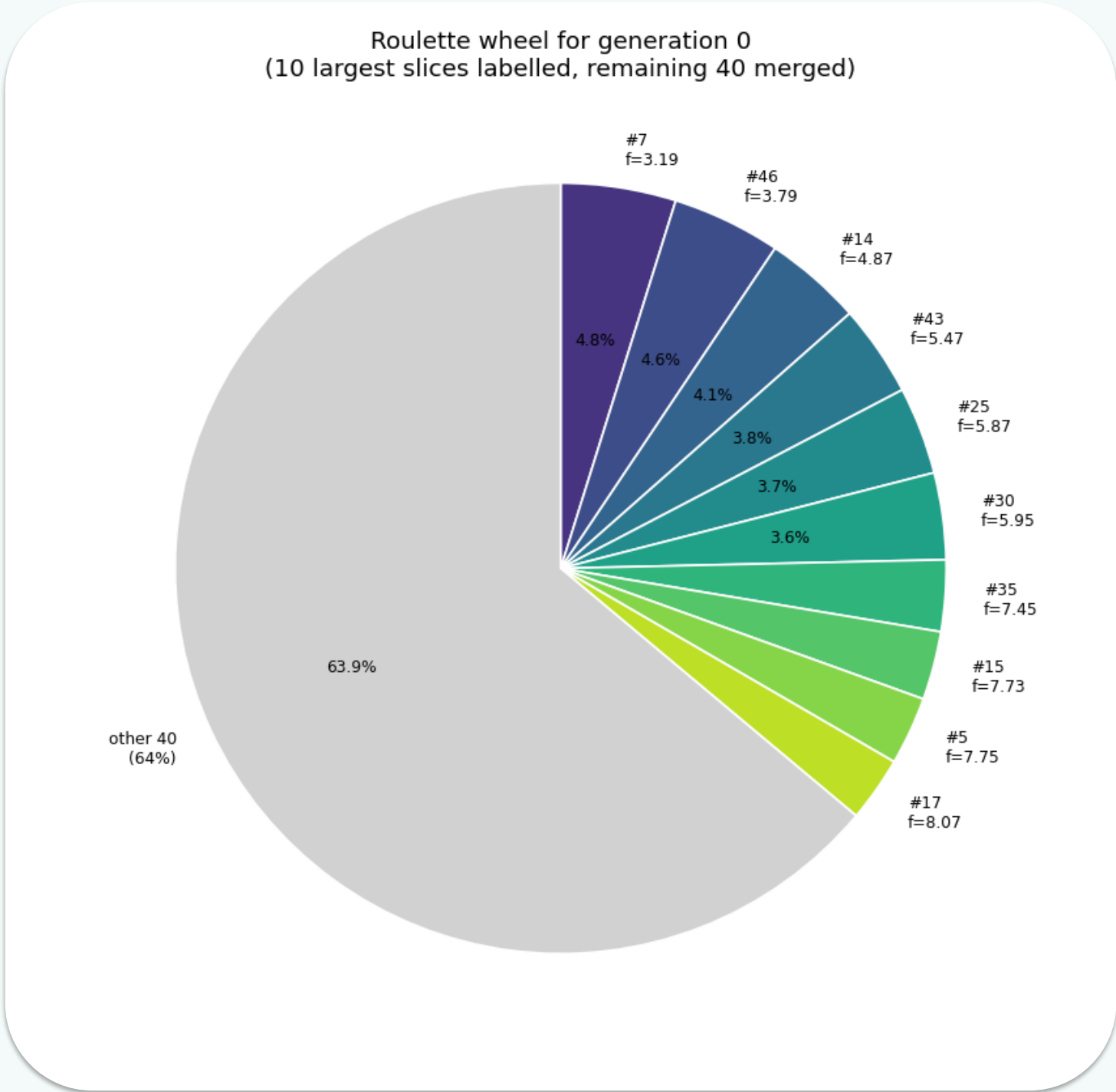
50 parents + 50 children → preserve the elite and fill the next population of 50.

# Better points get larger slices

Lower objective → higher fitness

$$F_i = (f_{\max} - f_i) + \delta, \quad p_i = \frac{F_i}{\sum_j F_j}$$
$$\delta = 0.10(f_{\max} - f_{\min}) + 10^{-12}$$

| Chromosome  | f     | Probability |
|-------------|-------|-------------|
| #7 (best)   | 3.19  | 4.83%       |
| #46         | 3.79  | 4.57%       |
| #14         | 4.87  | 4.10%       |
| #45 (worst) | 13.39 | 0.44%       |



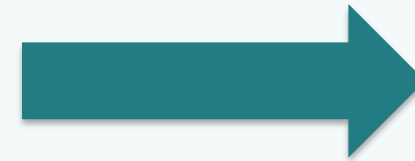
The code uses 50 probabilities. The chart groups the remaining 40 into one grey slice.

# Crossover swaps the second gene

PARENTS

X<sub>1</sub>X<sub>2</sub>

80%  
swap tails



CHILDREN



-4.0

-1.0

-1.5

2.5

20%: children are unchanged copies of the parents.

Only one legal cut exists. Crossover re-pairs values that are already in the population.

# Mutation creates new values

## 1 Check each gene

Each coordinate has its own 5% chance.

## 2 Add a random change

$$x_i \leftarrow x_i + N(0, \sigma^2)$$

## 3 Repair the boundary

Clip the changed coordinate to  $[-5, +5]$ .

ILLUSTRATION • NOT A RECORDED DRAW

**$[-1.5, -1.0]$**

↓ first gene changes

**$[-0.5, -1.0]$**

Boundary example:  $5.3 \rightarrow 5.0$

The 5% probability stays fixed. The size of the change is controlled by  $\sigma$ .

# Let progress control the step size

## Timed schedule

Steps shrink even when the run is still stuck.

## Our adaptive rule

$$\sigma = \text{clip}(f\_best / 4, 10^{-12}, 2)$$

| Best value f | Step size $\sigma$   |
|--------------|----------------------|
| 5            | 1.25                 |
| 2.6          | 0.65                 |
| $10^{-3}$    | $2.5 \times 10^{-4}$ |
| $10^{-12}$   | $10^{-12}$ (floor)   |

Near the origin:  $f \approx 4s$

$$s = \sqrt{(x_1^2 + x_2^2)/2}$$

$f/4$  estimates a local coordinate scale for Ackley; it is not the exact distance to the answer.

# Keep the best. Limit exact copies.

**50**

parents

+

**50**

children

=

**100**

candidates

**1****Keep the elite**

Best parent is protected.

**49 more seats**

Best remaining candidates, with  $\leq 15$  exact copies.  
If needed, fill empty seats with fresh random points.

Elitism protects the best-so-far value. The copy limit keeps more distinct candidates.

# Stop after 100 rounds—or little progress

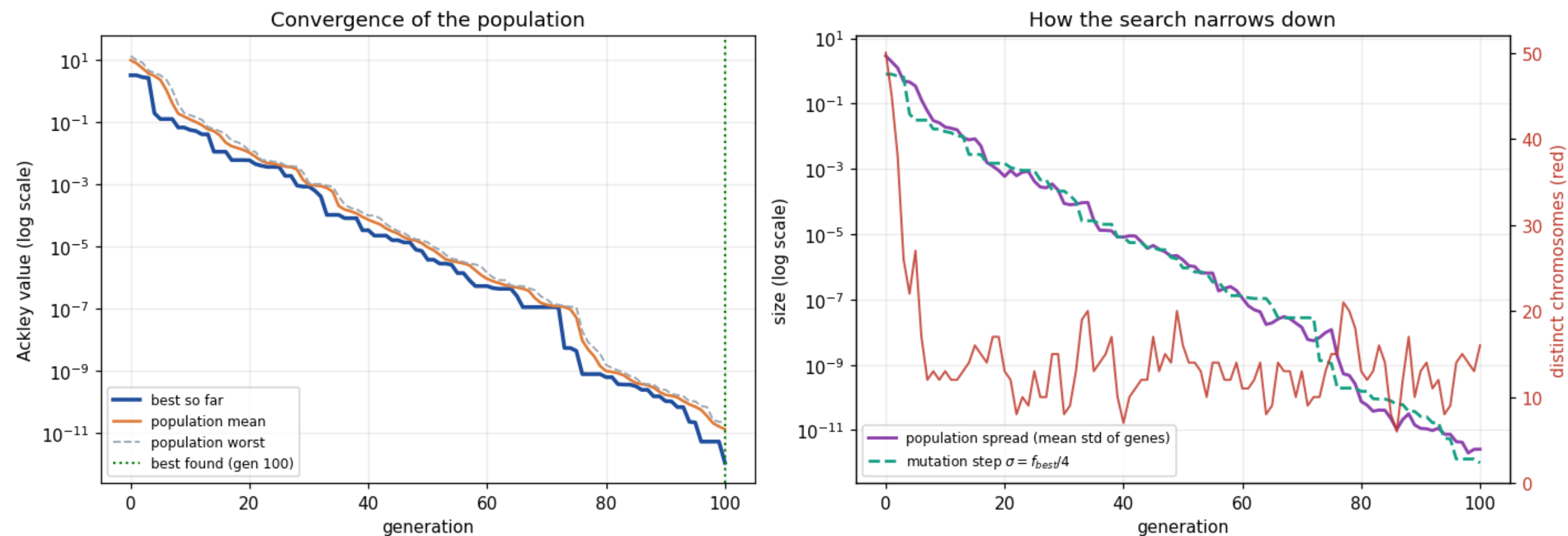
## 100 generations

maximum number of rounds

OR

## 20 small improvements in a row

each improvement  $< 10^{-10}$



Note: the plotted  $\sigma$  label is shorthand; the code clips  $\sigma$  to  $[10^{-12}, 2]$ .

**Seed 7:  $f \approx 0.00587$  at generation 20; the best result arrives at generation 100.**

# Our result is very close to (0, 0)

**$1.088 \times 10^{-12}$**

final objective value · seed 7

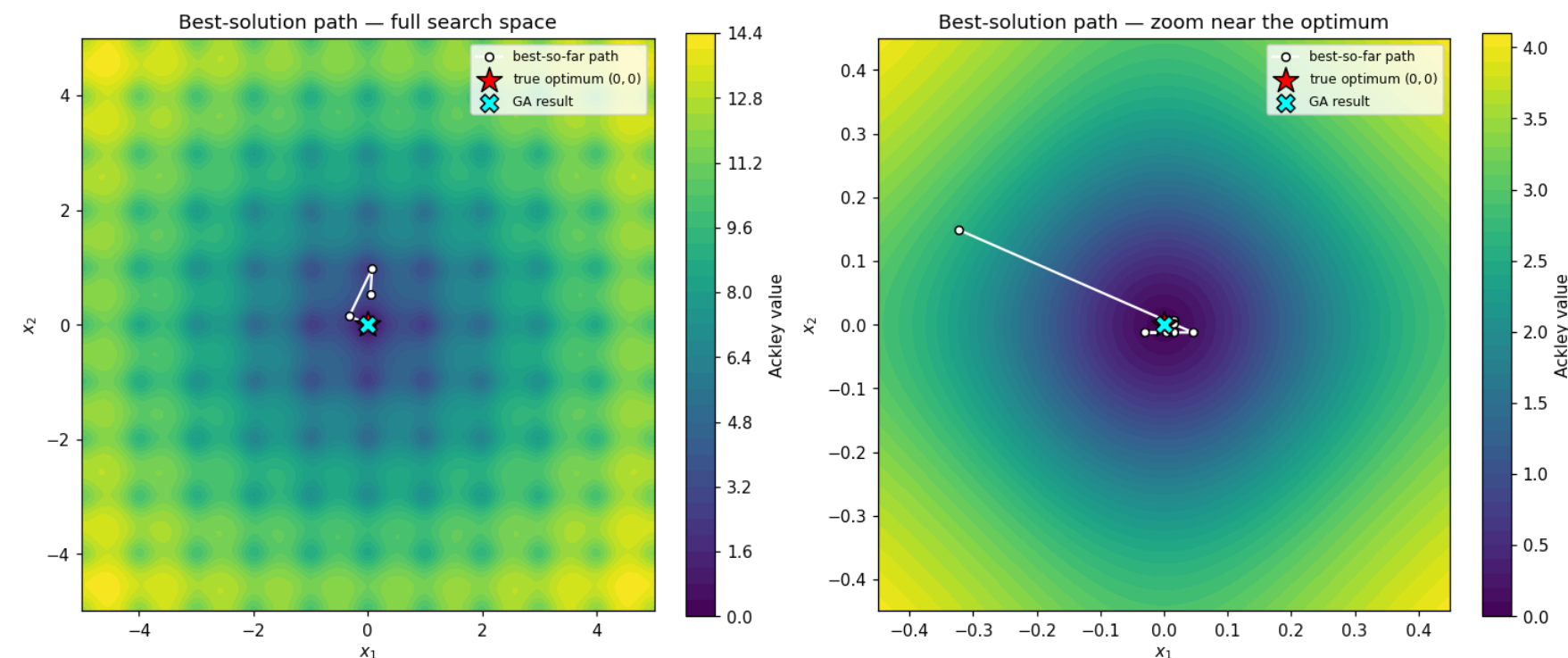
**100**

generation of the best result

**5,050**

function evaluations

$x_1 = -3.436 \times 10^{-13}$     $x_2 = -1.730 \times 10^{-13}$    distance =  $3.847 \times 10^{-13}$



Both coordinates have absolute error  $< 10^{-12}$ . The result is approximate.

**Random search with the same seed and 5,050 evaluations:  $f = 0.7693$ . This is one comparison.**

# Two changes solve different problems

Same 40 seeds in every row

| Step rule | Copy cap | Median f               | Worst f                | In valley | Distinct |
|-----------|----------|------------------------|------------------------|-----------|----------|
| Timed     | None     | $5.52 \times 10^{-9}$  | 2.58                   | 39 / 40   | 3        |
| Timed     | 15       | $6.76 \times 10^{-9}$  | 2.58                   | 39 / 40   | 9        |
| Adaptive  | None     | $5.00 \times 10^{-13}$ | $2.31 \times 10^{-11}$ | 40 / 40   | 8        |
| Adaptive  | 15       | $1.64 \times 10^{-12}$ | $3.14 \times 10^{-11}$ | 40 / 40   | 13       |

In valley: distance to (0, 0) < 0.5. Distinct: median number of unique chromosomes over the second half of each run, then the median across runs.

Adaptive step → smaller errors

Copy limit → more distinct points

# All 40 runs met the accuracy target

## 40 / 40

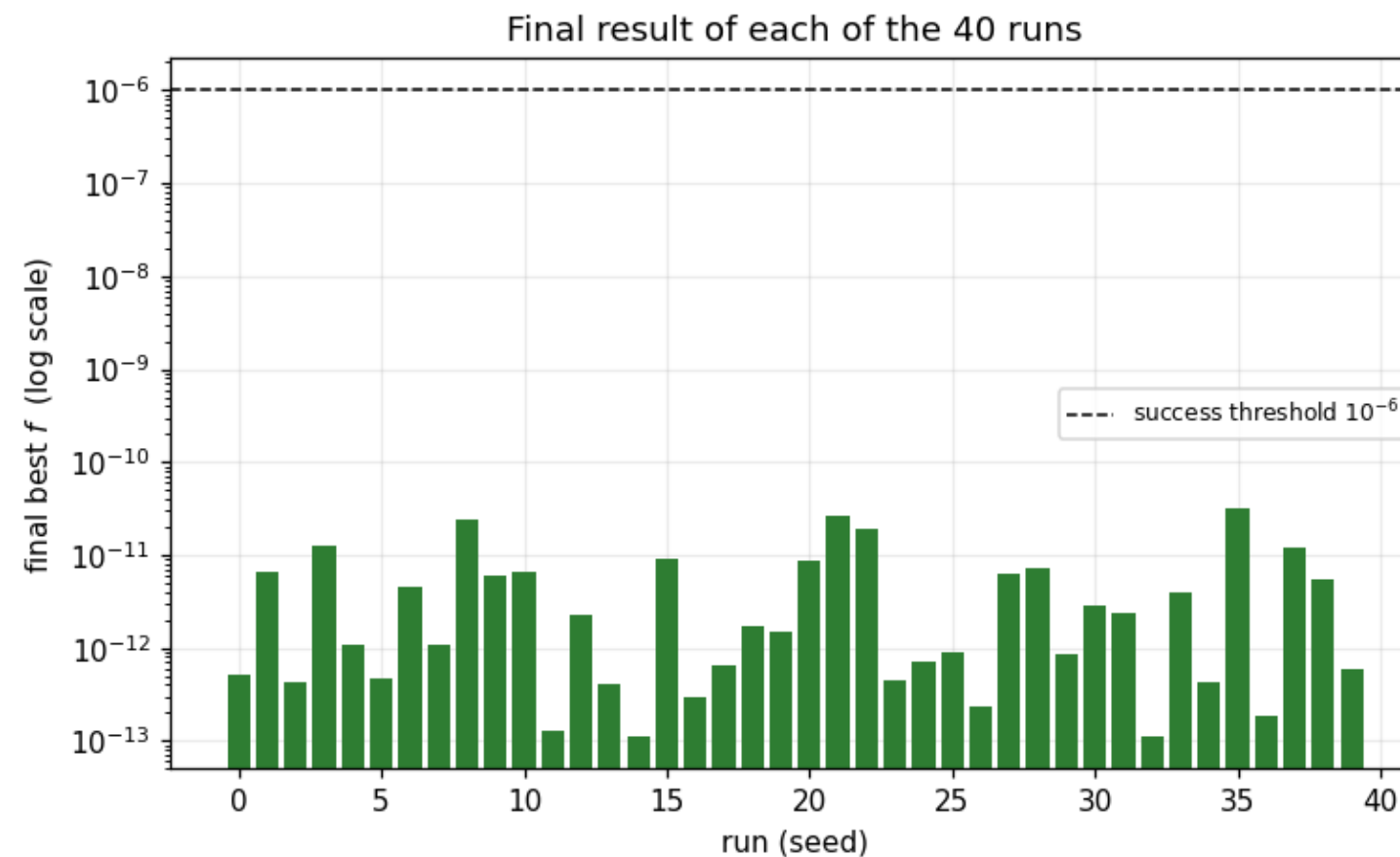
passed  $f < 10^{-6}$  · seeds 0–39

## $1.636 \times 10^{-12}$

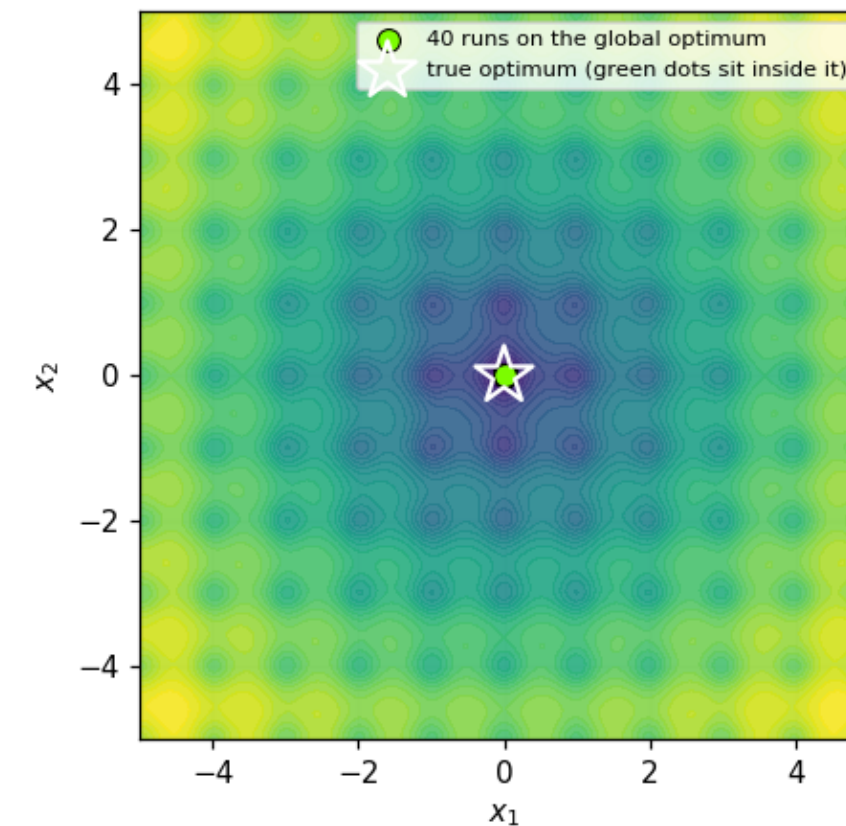
median final value

## $3.137 \times 10^{-11}$

worst final value



Where the 40 runs finished, on the real landscape



Best  $f = 1.106 \times 10^{-13}$ . Points visually overlap the optimum; they are approximate.

**All tested seeds met the target. This is evidence of reliability, not a guarantee for every run.**

# What we learned

**A good result needs  
a well-tested search.**

## Select + mix

Better parents; useful coordinate combinations.

## Preserve + diversify

Keep the best; limit exact copies.

## Mutate + adapt

New values; steps that shrink with progress.

## Check many seeds

40/40 below the target in our tests.



# Thank you. Questions?